

1 Chapter 1

Nick's Question 1.1 Proof of lemma 1.8.4

The proof is very nice.

2 Chapter 2

Nick's Question 2.1 In the proof of Theorem 2.4.3 on the bottom of page 34 there are some claims made that I am unable to verify why they are true. First, we are considering

$$X_t = X_0 + \int_0^t u(s)dB_s$$

and here $u(s)$ only satisfies

$$\mathbb{P}\left(\int_0^t u(s)^2 ds < \infty\right) = 1$$

so, as in the Kuo text, this stochastic integral exists and a limit in probability of a sequence of stochastic integrals $\int_0^t u_n(s)dB_s$ where the u_n are simple and are in L^2_{ad} and hence $\int_0^t u_n(s)dB_s$ is a martingale and we know the first and second moment. But for the limit $\int_0^t u(s)dB_s$ we know nothing about the moments. But in this text, it is claimed that

$$\mathbb{E}\left[\int_{t_i}^{t_{i+1}} u^2(s)ds - \left(\int_{t_i}^{t_{i+1}} u(s)dB_s\right)^2 \middle| \mathcal{F}_i\right] = 0$$

and I have never seen this claimed in any other text.

Note that u is bounded (we are assuming localization has been done). Hence, all moments exists.

Nick's Question 2.2 In the proof of theorem 2.5.1, why is it true that

$$E\left(\int_0^t |u'_n(B_s) - 1_{(x,\infty)}(B_s)|^2 ds\right) \leq \int_0^t \mathbb{P}\left(|B_s - x| \leq \frac{2}{n}\right) ds$$

Please explore the mollifier f and f_n . Plot these u'_n and see how could it differ from $1_{(x,\infty)}$.

Nick's Question 2.3 What is the motivation on page 41 behind the following definition of the backward ito integral

$$\int_0^T u(s)\hat{d}B_s = \lim_{n \rightarrow \infty} \sum_{i=0}^{n-2} \left(\frac{n}{T} \int_{(j+1)T/n}^{(j+2)T/n} u(s)ds\right) (B_{(j+1)T/n} - B_{jT/n})$$

The integral in the parenthesis is the average of u_s on $((j+1)T/n, (j+2)T/n)$.

Let's skip this part. I know nothing on the backward SDE and not particular interested either. It has applications in finance. If you don't do finance, I don't see the point here.

Nick's Question 2.4 (Problem 2.7)

This problem follows easily from Levy's Characterization of Brownian Motion. But I can't figure out how to do this problem with the hint given.

This is a nice problem. The hints is to ask you to find the distribution of $X_t - X_s$ through the characteristic function. Once apply Ito's formula, take expectation and remove the martingale part, you should arrive at an integral equation. Take derivatives, you arrive at a differential equation. The solution for $a_s(t) := \mathbb{E}(e^{i\lambda(X_t - X_s)})$ is

$$a_s(t) = e^{-\frac{\lambda^2}{2}(t-s)}.$$

Nick's Question 2.5 (Problem 2.15)

First we consider the 2 - d Ito process

$$\begin{pmatrix} M_s \\ N_s \end{pmatrix} = \begin{pmatrix} E(X^p) \\ E(Y^{q'}) \end{pmatrix} + \int_0^t \begin{pmatrix} \varphi_s & 0 \\ 0 & \psi_s \end{pmatrix} \begin{pmatrix} d\hat{B}_s \\ dB_s \end{pmatrix}$$

and then following the hint we apply Ito's formula to $f(x, y) = x^{1/p}y^{1/q'}$. We get the following derivatives:

- $f_x = \frac{1}{p}x^{\frac{1}{p}-1}y^{\frac{1}{q'}}$
- $f_{xx} = \frac{1}{p}\left(\frac{1}{p}-1\right)x^{\frac{1}{p}-2}y^{\frac{1}{q'}}$
- $f_y = \frac{1}{q'}x^{\frac{1}{p}}y^{\frac{1}{q'}-1}$
- $f_{yy} = \frac{1}{q'}\left(\frac{1}{q'}-1\right)x^{\frac{1}{p}}y^{\frac{1}{q'}-2}$

Now putting this into Ito's Formula gives us

$$\begin{aligned} M_t^{1/p} N_t^{1/q'} &= \|X\|_p \|Y\|_{q'} + \frac{1}{p} \int_0^t M_s^{\frac{1}{p}-1} N_s^{\frac{1}{q'}} \varphi_s d\hat{B}_s + \frac{1}{q'} \int_0^t M_s^{\frac{1}{p}} N_s^{\frac{1}{q'}-1} \psi_s dB_s \\ &\quad + \frac{1}{2} \frac{1}{p} \left(\frac{1}{p}-1\right) \int_0^t M_s^{\frac{1}{p}-2} N_s^{\frac{1}{q'}} \varphi_s^2 ds + \frac{1}{2} \frac{1}{q'} \left(\frac{1}{q'}-1\right) \int_0^t M_s^{\frac{1}{p}} N_s^{\frac{1}{q'}-2} \psi_s^2 ds \end{aligned}$$

I am not really sure where to continue from here. I see that if we let $t \rightarrow \infty$ then $M_t^{1/p} N_t^{1/q'} \rightarrow XY$ so if we took the expected value of both sides then we would partially have what we are looking for since $E(XY)$ is on the left and $\|X\|_p \|Y\|_{q'}$ is on the right. But I am not sure about the other terms. Im guessing the Ito integrals in the right have 0 expectation but I am not sure if that is in fact true since I dont see how it is easily seen that the integrands are in $L^2(\mathcal{P})$.

What I did next was see what $\hat{B}_s$ was equal too and for this we define another Ito Process

$$\begin{pmatrix} B_s \\ \tilde{B}_s \end{pmatrix} = \int_0^t \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} dB_s \\ d\tilde{B}_s \end{pmatrix}$$

and applied Itos formula to $f(t, x, y) = e^{-t}x + \sqrt{1 - e^{-2t}}y$ which gives us

$$\hat{B}_s = e^{-t}B_s + \sqrt{1 - e^{-2t}}\tilde{B}_s = \int_0^t \frac{e^{-2s}}{\sqrt{1 - e^{-2s}}}\tilde{B}_s - e^{-s}B_s ds + \int_0^t e^{-s}dB_s + \int_0^t \sqrt{1 - e^{-2s}}d\tilde{B}_s$$

However I am not sure if this step is necessary.

The followings are some of my thoughts: First of all, we need the absolute values:

$$\begin{pmatrix} M_s \\ N_s \end{pmatrix} = \begin{pmatrix} \mathbb{E}(|X|^p) \\ \mathbb{E}(|Y|^{q'}) \end{pmatrix} + \int_0^t \begin{pmatrix} \varphi_s & 0 \\ 0 & \psi_s \end{pmatrix} \begin{pmatrix} d\hat{B}_s \\ dB_s \end{pmatrix}.$$

Second, you forget

$$f_{xy} = \frac{1}{p} \frac{1}{q'} x^{-1/(p')} y^{-1/q'}.$$

Hence, by Ito's formula, we will have

$$\begin{aligned} M_t^{1/p} N_t^{1/q'} &= \|X\|_p \|Y\|_{q'} + \frac{1}{p} \int_0^t M_s^{\frac{1}{p}-1} N_s^{\frac{1}{q'}} \varphi_s d\hat{B}_s + \frac{1}{q'} \int_0^t M_s^{\frac{1}{p}} N_s^{\frac{1}{q'}-1} \psi_s dB_s \\ &\quad + \frac{1}{2} \frac{1}{p} \left(\frac{1}{p} - 1 \right) \int_0^t M_s^{\frac{1}{p}-2} N_s^{\frac{1}{q'}} \varphi_s^2 ds + \frac{1}{2} \frac{1}{q'} \left(\frac{1}{q'} - 1 \right) \int_0^t M_s^{\frac{1}{p}} N_s^{\frac{1}{q'}-2} \psi_s^2 ds \\ &\quad + \frac{1}{pq'} \int_0^t M_s^{-\frac{1}{p'}} N_s^{-\frac{1}{q}} \psi_s \phi_s d\langle B, \hat{B} \rangle_s \\ &=: \|X\|_p \|Y\|_{q'} + \sum_{i=1}^5 I_i. \end{aligned}$$

Upon taking expectation on both sides, we see that

$$LHS = \mathbb{E}(|X_t Y_t|) \geq \mathbb{E}(X_t Y_t).$$

We only need to prove that

$$\mathbb{E} \left(\sum_{i=1}^5 I_i \right) \leq 0,$$

which is clear for $i = 1, 2$, in which case we have the expectation equal to zero. Now for the rest three parts, we see that

$$I_3 + I_4 + I_5 = -\frac{1}{2pq'} M_s^{\frac{1}{p}-2} N_s^{\frac{1}{q'}-2} \left(N_s^2 \varphi_s^2 \frac{q'}{p'} - 2M_s N_s \phi_s \psi_s e^{-s} + M_s \psi_s^2 \frac{p}{q} \right),$$

where we have used the fact that $d\langle B, \hat{B} \rangle_s = e^{-s} ds$. Hence, the condition become the condition that the quadratic form has no solution:

$$\Delta = 4e^{-2s} - 4\frac{q'p}{p'q} = 4 \left(e^{-2s} - \frac{p-1}{q-1} \right) \leq 0$$

which is equivalent to

$$q \leq e^{2s}(p-1) + 1.$$

Hence, the largest q that makes the inequality work is $q = e^{2s}(p-1) + 1$.

3 Chapter 3

3.1 Reading Material

Nick's Question 3.1 (Smooth Cylindrical Random Variable) Following chapter 3 of the Kuo text, denote C the Wiener space of continuous functions from $[0,1]$ to $\mathbb{R}$, $\mathcal{R}$ the collection cylindrical sets

$$A = \{\omega \in C : (\omega(t_1), \dots, \omega(t_n)) \in U \mid t_i \in [0, 1], n \in \mathbb{N}, U \in \mathcal{B}(\mathbb{R}^n)\}$$

and μ the associated Wiener measure. Consider the coordinate processes $B(t, \omega) = \omega(t)$ which are Brownian motions under $(C, \mathcal{B}(\mathcal{R}), \mu)$. There is a theorem that can be found [here](#).

Theorem : The random variables

$$\{f(B(t_1, \cdot), \dots, B(t_n, \cdot)) \mid n \in \mathbb{N}, f \in \mathcal{S}(\mathbb{R}^n)\}$$

are dense in $L^2(C, \mu)$ where $\mathcal{S}(\mathbb{R}^n)$ are the Schwartz functions.

How does this relate to $\mathcal{S}$ defined on page 51 of the intro to Malliavin textbook? How are the cylindrical random variables defined in this case? Why is $\mathcal{S}$ dense in $L^2(\Omega)$? Even in Diffusions Markov Processes and Martingales - Roger and Williams, they define the cylindrical sigma field in the same way Kuo does- through the use of coordinate processes.

Nick Solution.

density

Theorem 3.1. Consider the probability space $(\Omega, \mathcal{F}, P)$ where $\mathcal{F}$ is the standard Brownian filtration. Then $\{f(B(h_1), \dots, f(B(h_n)), n \in \mathbb{N}, f \in C_p^\infty(\mathbb{R}^n), h_i \in H := L^2(\mathbb{R}^+)\}$ is dense in $L^2(\Omega)$.

Proof: This proof follows the same scheme as the solution found [here](#) which utilizes Dynkin's multiplicative system theorem. First, we can easily show that the sigma field generated by $\{B(h)\}_{h \in H}$ generates $\mathcal{F}$. This is true since $1_{[0,t]} \in H$ and $B(1_{[0,t]}) = B_t$. Second, since $\mathcal{S}(\mathbb{R}) \subset C_p^\infty(\mathbb{R})$ and since for every $h \in H$ there exists functions $f_n \in \mathcal{S}(\mathbb{R})$ such that $f_n \rightarrow h$ in $L^2(\mathbb{R})$ then by the Wiener Isometry we have $E|\int_0^\infty h(s) - f_n(s)dB_s|^2 = \int_0^\infty |h(s) - f_n(s)|^2 ds \rightarrow 0$. This implies that the sigma field generated by $\{f(B(h))\}$ where $h \in H$ and $f \in C_p^\infty(\mathbb{R})$ is the same as $\mathcal{F}$.

Now we let $M = \{f(B(h_1), \dots, B(h_n))\}$. We have already shown that $\sigma(M)$ is the same as $\mathcal{F}$. Also, M is closed under pointwise multiplication. Let $E := \bar{M}$ where closure is with respect to the $L^2(\Omega)$ norm. Let $H := E \cap L^\infty(\Omega)$. We have $M \subset H$. It can be shown that H is closed with respect to bounded pointwise convergence and also that H contains all constant functions. Thus, the hypothesis is Dynkin's multiplicative theorem has been satisfied and this implies that H contains all $\sigma(M) = \mathcal{F}$ measurable functions. But this means that it contains all bounded Borel measurable functions. But by the definition of H , this implies that E contains all the bounded Borel measurable functions. We know that all $X \in L^2(\Omega)$ can be approximated in the $L^2(\Omega)$ norm by bounded measurable functions. Also since E is closed and contains the bounded measurable functions, then E itself must be equal to $L^2(\Omega)$. Therefore we conclude that $\bar{M} = E = L^2(\Omega)$. □

Nick's Question 3.2 Why is $(D_t F)_{t \geq 0}$ a stochastic processes in $L^2(\Omega; H)$ where $H = L^2(\mathbb{R}^+)$? $D_t F$ is defined as

$$D_t F = \sum_{i=1}^n \frac{\partial f}{\partial x_i}(B(h_1), \dots, B(h_n)) h_i(t)$$

where $h_i \in H$ and $B(h_i) = \int_0^\infty h_i dB_s$ is the standard Wiener Integral. So isn't $D_t F$ real

valued since we are evaluating the $h_i(\cdot)$ at t ? Then shouldn't the $(D_t F)_{t \geq 0}$ be a $L^2(\Omega; \mathbb{R})$ valued stochastic process?

I believe that this is just a typo in the text

Nick's Question 3.3 (Proposition 3.2.4)(All questions have been solved with work below)

1. Why does $D_h(\delta(u)) = \langle u, h \rangle + \delta(D_h u)$? (I have answered this below)
2. Also, at the bottom of page 52, why is it that (I have answered this below)

$$E \left(\sum_{i,j=1}^{\infty} D_{e_i} \langle u, e_j \rangle D_{e_j} \langle v, e_i \rangle \right) = E \left(\int_0^{\infty} \int_0^{\infty} D_s u_t D_t v_s ds dt \right)$$

3. In the proof of property 3.3 why do we have (Answered below)

$$E \left(\sum_{i=1}^{\infty} \langle v, e_i \rangle \left(\langle u, e_i \rangle + \delta(D_{e_i}(u)) \right) \right) = E(\langle u, v \rangle) + E \left(\sum_{i,j=1}^{\infty} D_{e_i} \langle e, e_j \rangle D_{e_j} \langle v, e_i \rangle \right)$$

Proof of $D_h(\delta(u)) = \langle u, h \rangle + \delta(D_h u)$.

Since $u = \sum F_j h_j$ we just apply the definition of δ to get

$$\begin{aligned} D_h(\delta(u)) &= D_h \left(\sum_{j=1}^n F_j B(h_j) - \sum_{j=1}^n \langle DF_j, h_j \rangle \right) \\ &= \sum_{j=1}^n D_h(F_j B(h_j)) - \sum_{j=1}^n D_h \langle DF_j, h_j \rangle \\ &= \sum_{j=1}^n F_j \langle h, h_j \rangle + \sum_{j=1}^n B(h_j) D_h(F_j) - \sum_{j=1}^n D_h \langle DF_j, h_j \rangle \end{aligned} \quad (3.1) \quad \boxed{\text{P:3.2.4-1}}$$

where the third inequality follows from the product rule and $D_h(B(h_j)) = \langle h, h_j \rangle$. Now we focus on the last term from the last line above (3.1).

$$\sum_{j=1}^n D_h \langle DF_j, h_j \rangle = \sum_{j=1}^n \langle D_t \langle DF_j, h_j \rangle, h(t) \rangle \quad (3.2) \quad \boxed{\text{P:3.2.4-6}}$$

Before we continue, we simplify $D_t \langle DF_j, h_j \rangle$. We may assume that $F_j = f^j(B(h_1^j(t)), \dots, B(h_n^j(t)))$ where $f^j \in C_p^\infty(\mathbb{R}^n)$ and $h_i^j \in H$.

$$\begin{aligned} D_t \langle DF_j, h_j \rangle &= D_t \left\langle \sum_{i=1}^n \frac{\partial f^j}{\partial x_i} h_i^j(s), h_j(s) \right\rangle \\ &= \sum_{i=1}^n \langle h_i^j(s), h_j(s) \rangle D_t \left(\frac{\partial f^j}{\partial x_i} \right) \end{aligned} \quad (3.3) \quad \boxed{\text{P:3.2.4-2}}$$

We now plug (3.3) into (3.2) above to get

$$\sum_{j=1}^n \langle D_t \langle DF_j, h_j \rangle, h(t) \rangle = \sum_{j=1}^n \sum_{i=1}^n \langle h_i^j(s), h_j(s) \rangle \langle D_t \left(\frac{\partial f^j}{\partial x_i} \right), h(t) \rangle$$

$$\begin{aligned}
&= \sum_{j=1}^n \sum_{i=1}^n \left\langle h_i^j(s) \left\langle D_t \left(\frac{\partial f^j}{\partial x_i} \right), h(t) \right\rangle, h_j(s) \right\rangle \\
&= \sum_{j=1}^n \left\langle \left\langle D_t \left(\sum_{i=1}^n h_i^j(s) \frac{\partial f^j}{\partial x_i} \right), h(t) \right\rangle, h_j(s) \right\rangle \\
&= \sum_{j=1}^n \left\langle \left\langle D_t (DF_j), h(t) \right\rangle, h_j(s) \right\rangle \\
&= \sum_{j=1}^n \left\langle D_h (DF_j), h_j(s) \right\rangle
\end{aligned}$$

and if I haven't made a mistake, then what we have shown so far is that

$$\begin{aligned}
D_h(\delta(u)) &= \sum_{j=1}^n F_j \langle h, h_j \rangle + \sum_{j=1}^n B(h_j) D_h(F_j) - \sum_{j=1}^n D_h \langle DF_j, h_j \rangle \\
&= \sum_{j=1}^n F_j \langle h, h_j \rangle + \sum_{j=1}^n B(h_j) D_h(F_j) - \sum_{j=1}^n \left\langle D_h (DF_j), h_j \right\rangle
\end{aligned} \tag{3.4} \quad \boxed{\text{P:3.2.4-3}}$$

Also, the above is exactly line two of the first centered equations on page 53.

Now we need to show that the last two terms add up to $\delta(D_h(u))$. This is not defined in the text but I think that we need to define $D_h u$ as the following

$$D_h u = D_h \sum_{j=1}^n F_j h_j(t) = \sum_{j=1}^n h_j(t) D_h(F_j) \tag{3.5} \quad \boxed{\text{P:3.2.4-4}}$$

Thus now $D_h(u)$ (3.5) is of the form $\sum F_j h_j$ and so we may apply the definition of δ to it to get the following

$$\delta(D_h u) = \sum_{j=1}^n B(h_j) D_h(F_j) - \sum_{j=1}^n \langle D_s(D_h F_j), h_j(s) \rangle \tag{3.6} \quad \boxed{\text{P:3.2.4-5}}$$

Now we focus in the last term of the right hand side of (3.6). From (3.4) we note that we want to show that

$$\sum_{j=1}^n \langle D_s(D_h F_j), h_j(s) \rangle = \sum_{j=1}^n \left\langle D_h (DF_j), h_j \right\rangle \tag{3.7} \quad \boxed{\text{P:3.2.4-7}}$$

Here are the steps,

$$\begin{aligned}
\sum_{j=1}^n \langle D_s(D_h F_j), h_j(s) \rangle &= \sum_{j=1}^n \left\langle D_s \left(\left\langle \sum_{i=1}^n \frac{\partial f^j}{\partial x_i} h_i^j(t), h(t) \right\rangle \right), h_j(s) \right\rangle \\
&= \sum_{j=1}^n \left\langle D_s \left(\sum_{i=1}^n \frac{\partial f^j}{\partial x_i} \langle h_i^j(t), h(t) \rangle \right), h_j(s) \right\rangle \\
&= \sum_{j=1}^n \left\langle \sum_{i=1}^n D_s \left(\frac{\partial f^j}{\partial x_i} \right) \langle h_i^j(t), h(t) \rangle, h_j(s) \right\rangle \\
&= \sum_{j=1}^n \left\langle \left\langle \sum_{i=1}^n D_s \left(\frac{\partial f^j}{\partial x_i} \right) h_i^j(t), h(t) \right\rangle, h_j(s) \right\rangle
\end{aligned}$$

$$\begin{aligned}
&= \sum_{j=1}^n \left\langle \left\langle \sum_{i=1}^n D_s \left(\frac{\partial f^j}{\partial x_i} h_i^j(t) \right), h(t) \right\rangle, h_j(s) \right\rangle \\
&= \sum_{j=1}^n \left\langle \left\langle D_s(D_t F_j), h(t) \right\rangle, h_j(s) \right\rangle \\
&= \sum_{j=1}^n \left\langle D_h(D F_j), h_j \right\rangle
\end{aligned}$$

Hence (3.7) is shown and we are done.

Proof of bottom of page 52

$$\begin{aligned}
\sum_{i,j=1}^{\infty} D_{e_i} \langle u, e_j \rangle D_{e_j} \langle v, e_i \rangle &= \sum_{i,j=1}^{\infty} \left\langle D_s \langle u, e_j \rangle, e_i(s) \right\rangle \left\langle D_t \langle v, e_i \rangle, e_j(t) \right\rangle \\
&= \sum_{i,j=1}^{\infty} \left\langle \left\langle D_s(u_t), e_j(t) \right\rangle, e_i(s) \right\rangle \left\langle \left\langle D_t(v_s), e_i(s) \right\rangle, e_j(t) \right\rangle \\
&= \sum_{i=1}^{\infty} \left\langle \sum_{j=1}^{\infty} \left(\left\langle D_s(u_t), e_j(t) \right\rangle \left\langle \left\langle D_t(v_s), e_i(s) \right\rangle, e_j(t) \right\rangle \right), e_i(s) \right\rangle \\
&= \sum_{i=1}^{\infty} \left\langle \left\langle D_s(u_t), \left\langle D_t(v_s), e_i(s) \right\rangle \right\rangle, e_i(s) \right\rangle \\
&= \sum_{i=1}^{\infty} \left\langle \left\langle D_s(u_t), \left\langle D_t(v_q), e_i(q) \right\rangle \right\rangle, e_i(s) \right\rangle \\
&= \sum_{i=1}^{\infty} \int_0^{\infty} \left(\int_0^{\infty} D_s(u_t) \left(\int_0^{\infty} D_t(v_q) e_i(q) dq \right) dt \right) e_i(s) ds \\
&= \int_0^{\infty} \left(\int_0^{\infty} D_s(u_t) \left(\sum_{i=1}^{\infty} \int_0^{\infty} D_t(v_q) e_i(q) dq e_i(s) \right) dt \right) ds \\
&= \int_0^{\infty} \int_0^{\infty} D_s(u_t) D_t(v_s) dt ds
\end{aligned}$$

Where above we use the identity

$$\langle f, g \rangle = \sum_{i=1}^{\infty} \langle f, e_i \rangle \langle g, e_i \rangle.$$

Proof of step from property 3

$$\begin{aligned}
E \left(\sum_{i=1}^{\infty} \langle v, e_i \rangle \left(\langle u, e_i \rangle + \delta(D_{e_i}(u)) \right) \right) &= E(\langle u, v \rangle) + E \left(\sum_{i=1}^{\infty} \langle v, e_i \rangle \delta(D_{e_i}(u)) \right) \\
&= E(\langle u, v \rangle) + \sum_{i=1}^{\infty} E \left(\left\langle D(\langle v, e_i \rangle), D_{e_i}(u) \right\rangle \right) \quad (3.8) \quad \boxed{\mathbf{s-1}} \\
&= E(\langle u, v \rangle) + \sum_{i=1}^{\infty} E \left(\left\langle \left\langle D_s(v_t), e_i(t) \right\rangle, \left\langle D_t(u_s), e_i(t) \right\rangle \right\rangle \right) \\
&= E(\langle u, v \rangle) + \sum_{i,j=1}^{\infty} E \left(\left\langle \left\langle D_s(v_t), e_i(t) \right\rangle, e_j(s) \right\rangle \left\langle \left\langle D_t(u_s), e_i(t) \right\rangle, e_j(s) \right\rangle \right)
\end{aligned}$$

$$\begin{aligned}
&= E(\langle u, v \rangle) + \sum_{i,j=1}^{\infty} E \left(D_{e_j} \langle v_t, e_i(t) \rangle \left\langle \langle D_t(u_s), e_j(s) \rangle, e_i(t) \right\rangle \right) \\
&= E(\langle u, v \rangle) + \sum_{i,j=1}^{\infty} E \left(D_{e_j} \langle v_t, e_i(t) \rangle D_{e_i} \langle u_s, e_j(s) \rangle \right) \\
&= E(\langle u, v \rangle) + \sum_{i,j=1}^{\infty} E \left(D_{e_j} \langle v, e_i \rangle D_{e_i} \langle u, e_j \rangle \right)
\end{aligned} \tag{3.9} \quad \boxed{\text{s-2}}$$

where in (3.8) we used Proposition 3.2.3 and in (3.9) we use Fubini's Theorem.

Nick's Question 3.4 (Why is δ linear?) On page 55 it says that δ is linear. First, for $u = \sum_{j=1}^n F_j h_j$ we have

$$\delta(u) = \sum_{j=1}^n F_j B(h_j) - \sum_{j=1}^n \langle DF_j, h_j \rangle_H$$

Now if we have a $v = \sum_{j=1}^n G_j h'_j$ how does $\delta(u+v) = \delta(u) + \delta(v)$. Also how could we write $\sum_{j=1}^n F_j h_j + \sum_{j=1}^n G_j h'_j = \sum_{j=1}^n K_j k_j$ where $K_j \in \mathcal{S}$ and $k_j \in H$

Do we do this through the relationship $E(\langle DF, u \rangle) = E(\delta(u)F)$ in the sense that

$$\begin{aligned}
E(\delta(u+v)F) &= E(\langle DF, u+v \rangle) \\
&= E(\langle DF, u \rangle) + E(\langle DF, v \rangle) \\
&= E(\delta(u)F) + E(\delta(v)F) \\
&= E((\delta(u) + \delta(v))F)
\end{aligned}$$

and if this is true for every F then I believe we can say $\delta(u+v) = \delta(u) + \delta(v)$? **Yes, this is the argument to show that δ is linear. In general, for two distributions F and G , how one identifies $F = G$? This is true iff $\langle F, \varphi \rangle = \langle G, \varphi \rangle$ for all test function φ . Here, is the same. In order to identify that two random variables $\delta(u+v)$ and $\delta(v) + \delta(u)$ are same, this relation has to be tested against all functions in its dual space, which is $\mathbb{D}^{1,2}$.**

Nick's Question 3.5 Exercise 3.1 I don't see how we can apply this hint? Suppose we are able to show that the hint is true then how does that show $\mathcal{P}$ is dense in $L^q(\Omega)$? There is a theorem that $\{x_n\}$ is a basis for a Hilbert space H if and only if $\sum \langle x, x_n \rangle x_n = 0$ implies $x = 0$ but this is only valid for Hilbert space and $L^q(\Omega)$ for $q \neq 2$ is not a Hilbert space since there is no inner product. **This is the standard argument to show one set is dense in another. You may miss the fact that the set of polynomial functions is dense in $C_0^\infty(\mathbb{R}^d)$, the latter is dense in the set of the Schwartz test function $\mathcal{S}(\mathbb{R}^d)$. Please do justify yourself this problem.**

Nick's Question 3.6 Prop 3.3.2 page 54 This relates to exercise 3. In the proof of prop 3.3.2 how can we use Holders inequality to show that $\varphi'(F)DF \in L^q(\Omega, H)$?

Nicks Solution.

$$[E \|\varphi'(F)DF\|_H^q]^{1/q} = \left[\int_{\Omega} \left(\int_0^\infty \varphi'(F)^2 (D_t F)^2 dt \right)^{q/2} dP \right]^{1/q}$$

$$\begin{aligned}
&\leq \left[\int_{\Omega} (1 + |F|^{\alpha})^q \left(\int_0^{\infty} (D_t F)^2 dt \right)^{q/2} dP \right]^{1/q} \\
&= \left\| (1 + |F|^{\alpha}) \left(\int_0^{\infty} (D_t F)^2 dt \right)^{1/2} \right\|_q
\end{aligned}$$

Note that $\frac{1}{q} = \frac{1}{p} + \frac{1}{p/\alpha}$ and so if we apply Holders Inequality above we get that

$$[E \|\varphi'(F)DF\|_H^q]^{1/q} \leq \|1 + |F|^{\alpha}\|_{p/\alpha} \left\| \left(\int_0^{\infty} (D_t F)^2 dt \right)^{1/2} \right\|_p$$

which is finite since $F \in \mathbb{D}^{1,p}$. Or equivalently,

$$\begin{aligned}
\|\varphi'(F)DF\|_H^2 &= \|\varphi(F)\|_q^2 \|DF\|_H^2 \\
&\leq \|\varphi(F)\|_{\frac{p}{\alpha}}^2 \|DF\|_p^2 \\
&= \|\varphi(F)\|_{\frac{p}{\alpha}}^2 \int_0^{\infty} \|D_t F\|_p^2 dt
\end{aligned}$$

□

Nick's Question 3.7 Prop 3.3.2 page 54

I have not been able to verify why $\varphi(F) \in \mathbb{D}^{1,q}$. By definition, this means that we need to show that there exists $X_n \in \mathcal{S}$ such that

1. $X_n \rightarrow \varphi(F)$ in $L^q(\Omega)$
2. $D(X_n) \rightarrow D(\varphi(F))$ in $L^q(\Omega, H)$

You have made the problem more complex than what it should be. You only need to check the seminorm $\|\cdot\|_{1,p}$ is finite.

By assumption we already have that there exists $\{F_n\} \subset \mathcal{S}$ such that $F_n \rightarrow F$ in $L^p(\Omega)$ and also $DF_n \rightarrow DF$ in $L^p(\Omega, H)$. We then take a sequence of mollifiers, $\{\alpha_n\}$ and consider the functions $\varphi_k(x) = (\varphi * \alpha_k)(x)$ which by the original assumption on φ are $\alpha + 1$ holder continuous. You cannot say a function is β -Hölder continuous with $\beta \geq 1$. That means the function is a constant.

Now, we notice that by properties of the mollifier since $\varphi(F) \in L^q(\Omega)$ then

$$E|\varphi_k(F_j) - \varphi(F_j)|^q \xrightarrow{j \rightarrow \infty} 0$$

Next, since φ is $\alpha + 1$ holder continuous then we have

$$\begin{aligned}
E|\varphi(F_j) - \varphi(F)|^q &\leq KE|F_j - F|^{q(\alpha+1)} \\
&= KE|F_j - F|^p \rightarrow 0
\end{aligned}$$

and thus we conclude as k and j go to infinity that $\varphi_k(F_j) \rightarrow \varphi(F)$ in $L^q(\Omega)$ and $\varphi_k(F_j) \in \mathcal{S}$.

I am unable to show why $D_t(\varphi_k(F_j)) \rightarrow D_t(\varphi(F))$ in $L^q(\Omega, H)$.

However since $F_j = f^j(B(h_1^j), \dots, B(h_n^j))$, we see that

$$D_t(\varphi_k(F_j)) = \sum_{i=1}^n \frac{\partial}{\partial x_i} (\varphi * \alpha_k)(F_j) h_i^j(t)$$

$$\begin{aligned}
&= \sum_{i=1}^n (\varphi' * \alpha_k)(F_j) \frac{\partial f^j}{\partial x_i} h_i^j(t) \\
&= (\varphi' * \alpha_k)(F_j) \sum_{i=1}^n \frac{\partial f^j}{\partial x_i} h_i^j(t) \\
&= (\varphi' * \alpha_k)(F_j) D_t(F_j).
\end{aligned}$$

We know that $\{\alpha_k\}$ are mollifiers so $(\varphi' * \alpha_k)(F_j) \rightarrow \varphi'(F_j)$ almost surely and by assumption $D_t(F_j) \rightarrow D_t(F)$ in $L^p(\Omega, H)$. But I still am having trouble rigorously showing that $D_t(\varphi_k(F_j)) \rightarrow D_t(\varphi(F_j))$ in $L^p(\Omega, H)$. **You have two indices, you need to take the limit once for just one indices, not the two at the same time. More precisely,**

$$\|D_t(\varphi_k(F_j) - D_t(\varphi(F)))\|_p \leq \|D_t(\varphi_k(F_j)) - D_t(\varphi(F_j))\|_p + \|D_t(\varphi(F_j)) - D_t(\varphi(F))\|_p.$$

Actually, you need to split to three such differences. I will explain to you during the meeting.

Nick's Question 3.8 Exercise 3.4 Let $M = \sum_{t \in [0,1]} X_t$. The hint says to approximate this by taking the sup over a finite set. We know that since X_t is continuous in t then $M = \sup_{t \in \mathbb{A}} X_t$ where $A = \mathbb{Q} \cap [0, 1]$. Let $\{a_n\}$ be an enumeration of $A \setminus \{0, 1\}$ and let $S_k = \{0, a_1, \dots, a_k, 1\}$ then each S_k are finite and $M_k := \sup_{t \in S_k} X_t$ is an **(Note how to type $\mathbb{Q}$, you need curly braces.)** approximation of M since $M_k \rightarrow M$ almost surely. The hint also tells us to use Corollary 4.2.5 but that requires us to find $D_t M_k$ which I am unsure of how to do. The hint says that we need to use exercise 3.3 which is chain rule for Lipschitz functions but I do not know how that applies. This corollary would also require us to show that $M_k \xrightarrow{L^2(\Omega)} M$ which I have been unable to show either. **Try harder... This is an interesting problem that is related the hitting problems that Robert Dalang and Eulalia have done a lot work in the SPDE setting.**

3.2 Exercises

Exercise 3.3 Let α_n be a sequence of mollifiers and $\varphi_k := (\varphi * \alpha_k)$. Since φ is Lipschitz with Lipschitz constant K then this implies that $\varphi'_k \leq K$ and in addition φ' exists almost everywhere and is also bounded by K . By proposition 3.3.2 we have that

$$D_t(\varphi_k(F)) = \varphi'_k(F) D_t(F)$$

Also note that

$$\begin{aligned}
|\varphi_k(F) - \varphi(F)|^2 &= \left| \int_{\mathbb{R}} (\varphi(F - y) - \varphi(F)) \alpha_k(y) dy \right|^2 \\
&\leq K \left| \int_{\mathbb{R}} y \alpha_k(y) dy \right|^2 \\
&\leq K'
\end{aligned}$$

Hence $|\varphi_k(F) - \varphi(F)|^2$ is bounded above and also converges to 0 almost surely so by the dominated convergence theorem we have that

$$\mathbb{E}|\varphi_k(F) - \varphi(F)|^2 \rightarrow 0$$

and so the hypothesis of corollary 4.2.5 is satisfied. This implies that $\varphi(F) \in \mathbb{D}^{1,2}$ and

$$D_t(\varphi_n(F)) \rightarrow D_t(\varphi(F))$$

weakly in $L^2(\Omega, H)$. Lastly, since $\varphi'_k(F)D_t(F)$ is bounded in $L^2(\Omega, H)$ and also

$$\varphi'_k(F)D_t(F) \rightarrow GD_t(F)$$

almost surely where G is a random variable bounded by K then

$$\varphi'_k(F)D_t(F) \rightarrow GD_t(F)$$

weakly in $L^2(\Omega, H)$ as well. Since weak limits are unique then we must have

$$D_t(\varphi(F)) = GD_t(F)$$

However note that in general we have as a property of mollifiers that $\varphi'_n(x) \rightarrow \varphi'(x)$ almost surely. But we said before that $\varphi'(x)$ only exists almost surely since φ is Lipschitz. Therefore if the law of F is absolutely continuous with respect to Lebesgue measure then we have $G = \varphi'(F)$ almost surely. Indeed, let $\mathcal{N}$ be the null set where φ' is undefined. If the law of F is absolutely continuous with respect to Lebesgue measure then $P(F \in \mathcal{N}) = 0$ and we must have $G = \varphi'(F)$ almost surely.

Exercise 3.4 I found a variant of this problem [here](#) where they consider $\{B_t\}_{t \in [0,1]}$ instead of an arbitrary centered Gaussian processes $\{X_t; t \in [0,1]\}$. They use the idea I was trying, which is we want to consider the function $f(x) = \max_{i \in \{1, \dots, n\}} x_i$ for $x \in \mathbb{R}^n$. f is a Lipschitz function and if we consider $\{q_n\}$ an enumeration of $\mathbb{Q} \cap [0,1]$ and consider $Y_n = (X_{q_1}, \dots, X_{q_n})$ then $M_n := \max_{i \in \{1, \dots, n\}} \{X_{q_1}, \dots, X_{q_n}\} = f(Y_n)$ and $M_n \rightarrow M$ almost surely. However in the above linked pdf they claim without proof that that $M_n \rightarrow M$ in $L^2(\Omega)$ which I have so far been unable to show.

I was unaware that there is an extension of the chain rule for Lipschitz functions that is given in the above linked pdf that says we may consider Lipschitz functions on $\mathbb{R}^d$ and random vectors $F = (F_1, \dots, F_n)$ whose components are in $\mathbb{D}^{1,2}$ and then $f(F)$ will also be in $\mathbb{D}^{1,2}$. We apply this above to see that $M_n := \max_{i \in \{1, \dots, n\}} \{X_{q_1}, \dots, X_{q_n}\} = f(Y_n) \in \mathbb{D}^{1,2}$. Next we define the sets

$$A_1 = \{M_n = X_{q_1}\}$$

...

$$A_k = \{M_n \neq X_{q_1}, M_n \neq X_{q_2}, \dots, M_n = X_{q_k}\}, 2 \leq k \leq n$$

and by doing so

$$\begin{aligned} DM_n &= D \left(\sum_{k=1}^n 1_{A_k} X_{q_k} \right) \\ &= \sum_{k=1}^n 1_{A_k} D(X_{q_k}) \end{aligned}$$

where the last equality is to follow because Lemma 3.4.2 on page 56 but I need to verify why. Lets suppose for now that $\{X_t\}$ are Brownian motions. Then

$$D(X_{q_k}) = 1_{[0, q_k]}(t) \tag{3.10} \quad \boxed{\text{brown}}$$

and the above becomes

$$\begin{aligned} DM_n &= \sum_{k=1}^n 1_{A_k} D(X_{q_k}) \\ &= \sum_{k=1}^n 1_{A_k} 1_{[0, q_k]}(t) \end{aligned}$$

and we have that

$$\begin{aligned}\mathbb{E} \|DM_n\|_H^2 &= \mathbb{E} \left\| \sum_{k=1}^n 1_{A_k} 1_{[0, q_k]}(t) \right\|_H^2 \\ &= \mathbb{E} \int \left(\sum_{k=1}^n 1_{A_k} 1_{[0, q_k]}(t) \right)^2 dt \\ &= \sum_{k=1}^n \mathbb{E} \left(1_{A_k} \int 1_{[0, q_k]}(t) dt \right)\end{aligned}$$

where the last inequality follows from the disjointness of the A_k . Continuing the calculations we see

$$\begin{aligned}\mathbb{E} \|DM_n\|_H^2 &= \sum_{k=1}^n \mathbb{E} \left(1_{A_k} \int 1_{[0, q_k]}(t) dt \right) \\ &= \sum_{k=1}^n q_k \mathbb{E} (1_{A_k}) \\ &\leq \sum_{k=1}^n \mathbb{E} (1_{A_k}) \\ &\leq \mathbb{E} (\Omega) = 1\end{aligned}$$

and so we apply Corollary 4.2.5 to conclude that $M = \sup_{t \in [0, 1]} X_t \in \mathbb{D}^{1,2}$ and that $DM_n \rightarrow DM$ in the weak topology of $L^2(\Omega, H)$ which means that for all $G \in \mathbb{D}^{1,2}$

$$\langle DM_n, G \rangle_{L^2(\Omega, H)} \rightarrow \langle DM, G \rangle_{L^2(\Omega, H)}$$

To prove that $DX = 1_{[0, T]}$ we note that $\sum_{k=1}^n 1_{A_k} q_k$ which converges almost surely to T . Then we can conclude that $DM_n = \sum_{k=1}^n 1_{A_k} 1_{[0, q_k]}(t)$ converges to $1_{[0, T]}$ almost surely. Since DM_n converges both weakly and almost surely then a result from functional analysis says that these limits must be equal and we see that $DM = 1_{[0, T]}$.

Now if we go back to (3.10) and assume that $\{X_t\}$ is a centered Gaussian process then I am unsure of how to continue. For example we can not calculate $D(X_t)$ for any $t \in [0, 1]$ unless we know that $X_t = \int_0^\infty h(s) dB_s$ for some $h \in H$ but I do not think that this can be claimed.

4 Chapter 4

4.1 Reading material

Nick's Question 4.1 proof of proposition 4.2.1 In the conclusion of this proof, Nualart refers to a density argument but what is he referring too? We see that from (4.1) on page 64 that

$$\mathbb{E} I_n^2(f) = n! \left\| \tilde{f} \right\|_{L^2(\mathbb{R}_+^n)}$$

so is there a result that says that functions

$$A := \{g^{\otimes n}(x_1, \dots, x_n) = g(x_1) \cdots g(x_n), g \in L^2(\mathbb{R}^+)\}$$

are dense in $L_s^2(\mathbb{R}_+^n)$? If that's the case then I believe that for a given $f \in L_s^2(\mathbb{R}_+^n)$ then we

find $\{g_k\} \in A$ such that

$$\mathbb{E}|I_n(f - g_k)|^2 = n! \left\| \tilde{f} - g_k \right\|_{L^2(\mathbb{R}_+^n)}^2 \rightarrow 0$$

and I think since D is a closed operator then we need

$$D_t(I_n(g_k)) = nI_{n-1}(g_k(\cdot, t)) \rightarrow D_t(I_n(f))?$$

Nick's Question 4.2 Example 4.2.3 We know that $B_1^3 = \sum_{n=0}^{\infty} I_n(f_n)$ and by the bottom of page 66, for this example we use the formula

$$f_n = \frac{1}{n!} \mathbb{E}(D^n B_1^3)$$

and calculating each of the above we see that

$$\begin{aligned} f_0 &= \mathbb{E}B_1^3 = 0 \\ f_1 &= 31_{[0,1]}(t_1) \\ f_2 &= 0 \\ f_3 &= 1_{[0,1]}(t_1)1_{[0,1]}(t_2)1_{[0,1]}(t_3) \\ f_k &= 0, \quad k \geq 4 \end{aligned}$$

thus we see that by using the definition on page 63 that

$$\begin{aligned} B_1^3 &= I_1(31_{[0,1]}(t_1)) + I_3(1_{[0,1]}(t_1)1_{[0,1]}(t_2)1_{[0,1]}(t_3)) \\ &= \int_0^\infty 31_{[0,1]}(t_1)dB_{t_1} + \int_0^\infty \int_0^{t_3} \int_0^{t_2} 1_{[0,1]}(t_1)1_{[0,1]}(t_2)1_{[0,1]}(t_3)dB_{t_1}dB_{t_2}dB_{t_3} \\ &= 3B_1 + \int_0^1 \int_0^{t_3} 1_{[0,1]}(t_2) \int_0^{t_2} 1_{[0,1]}(t_1)dB_{t_1}dB_{t_2}dB_{t_3} \\ &= 3B_1 + \int_0^1 \int_0^{t_3 \wedge 1} \int_0^{t_2 \wedge 1} dB_{t_1}dB_{t_2}dB_{t_3} \\ &= 3B_1 + \int_0^1 \int_0^{t_3} \int_0^{t_2} dB_{t_1}dB_{t_2}dB_{t_3} \end{aligned}$$

This is different from the answer in the test on page 67. Is mine correct?

Nick's Question 4.3 directional derivative page 69

1. $\tau_g(\omega) = \omega + g$ where g is continuous on $\mathbb{R}^+$ and $g(0) = 0$. Why does this imply that

$$\tau_g(B(h)) = B(h) + \langle h, \dot{g} \rangle_H.$$

when by definition

$$\tau_g(B(h)) = B(h) + g$$

but also $B(h) \notin \Omega$ where Ω is the Wiener space and

$$B(h) = \int_0^\infty h(s)dB_s$$

so why does $\tau_g(B(h))$ exist?

2. Why does Girsanov's theorem imply that if $F \in L^2(\Omega)$ then $F(\omega + g)$ is almost surely defined?
3. What is the motivation behind the definition of the directional derivative

$$\tilde{D}_g F = \lim_{\epsilon \rightarrow 0} \frac{1}{\epsilon} (F \circ \tau_{\epsilon g} - F)?$$

In this pdf they claim that when we have $DF = \sum_{i=1}^n \partial_i f(W(h_1), \dots, W(h_n)) h_i$ then the directional derivative is

$$\langle DF, h \rangle_H = \lim_{\epsilon \rightarrow 0} \frac{1}{\epsilon} (f(W(h_1) + \epsilon \langle h_1, h \rangle, \dots, W(h_n) - \epsilon \langle h_n, h \rangle) - F) \quad (4.1) \quad \text{pdf 1}$$

and also it claims that $\langle DF, h \rangle_H$ is the derivative at $\epsilon = 0$ of the shifted processes

$$F(W(g) + \epsilon \langle g, h \rangle) \quad (4.2) \quad \text{pdf 2}$$

for $g \in H$ but what does this mean?

Consider $F = (W(h_1), \dots, W(h_n))$ and here we have

$$DF = \sum_{i=1}^n \frac{\partial f}{\partial x_i} (W(h_1), \dots, W(h_n)) h_i$$

and so if we try to calculate $\langle DF, h \rangle$ we get

$$\begin{aligned} \langle DF, h \rangle &= \sum_{i=1}^n \frac{\partial f}{\partial x_i} (W(h_1), \dots, W(h_i), \dots, W(h_n)) \langle h_i, h \rangle \\ &= \sum_{i=1}^n \lim_{\epsilon \rightarrow 0} \frac{f(W(h_1), \dots, W(h_i) + \epsilon \langle h_i, h \rangle, \dots, W(h_n)) - F}{\epsilon \langle h_i, h \rangle} \langle h_i, h \rangle \\ &= \sum_{i=1}^n df \lim_{\epsilon \rightarrow 0} \frac{f(W(h_1), \dots, W(h_i) + \epsilon \langle h_i, h \rangle, \dots, W(h_n)) - F}{\epsilon} \end{aligned} \quad (4.3) \quad \text{ddiv}$$

but I do not see how we can simplify this to (4.1)?

Solution for 3 (4.1): The definition of the directional derivative is

$$D_v(f) = \lim_{h \rightarrow 0} \frac{f(x + hv) - f(x)}{h} = \frac{d}{d\epsilon} f(x + \epsilon v)|_{\epsilon=0}$$

and according to wikipedia, when the partials of f exist then it can be shown that

$$D_v(f) = \sum_{i=1}^n v_i \frac{\partial f}{\partial x_i} \quad (4.4) \quad \text{dirdiv}$$

where $v = (v_1, \dots, v_n)$. This is stated without proof and is supposed to follow from the definition of the total derivative which is something I am not familiar too. However, if we let $v = (\langle h_1, h \rangle, \dots, \langle h_n, h \rangle)$ then

$$\begin{aligned} D_v(f) &= \lim_{\epsilon \rightarrow 0} \frac{1}{\epsilon} (f(W(h_1) + \epsilon \langle h_1, h \rangle, \dots, W(h_n) - \epsilon \langle h_n, h \rangle) - F) \\ &= \sum_{i=1}^n \langle h_i, h \rangle \frac{\partial f}{\partial x_i} \end{aligned}$$

$$= \langle DF, h \rangle$$

where the second equality follows from (4.4) and the third equality follows from the work listed in the gray text from the question I asked.

Nick's Question 4.4 Proof of Theorem 4.4.2

1. Why does

$$\mathbb{E} \left(\left| \frac{1}{\epsilon} (F_n \circ \tau_{\epsilon g} - F_n) - \langle DF_n, \dot{g} \rangle_H \right| \right) \rightarrow 0$$

for fixed n and $\epsilon \rightarrow 0$?

I believe that they are claiming without proof that if $F_n \in \mathcal{S}$ then

$$\frac{1}{\epsilon} (F_n \circ \tau_{\epsilon g} - F_n) \rightarrow \langle DF_n, \dot{g} \rangle_H$$

I assume in $L^2(\Omega)$ as $\epsilon \rightarrow 0$.

I believe this is true in the most simple example where $F_n = \int_0^T h_n(s) dB_s$ where $h_n \in H$ and B_s is the canonical brownian motion on the Wiener space, ie $B_s(\omega) = \omega(s)$. For any $g \in \mathcal{H}$, or in other words

$$g(t) = \int_0^t \dot{g}(s) ds$$

then we can easily calculate the following derivative

$$\begin{aligned} \tilde{D}_g(F_n)(\omega) &= \lim_{\epsilon \rightarrow 0} \frac{1}{\epsilon} (F_n \circ \tau_{\epsilon g} - F_n) = \lim_{\epsilon \rightarrow 0} \frac{1}{\epsilon} \left(\int_0^t h_n(s) dB_s(\omega + \epsilon g) - \int_0^t h_n(s) dB_s(\omega) \right) \\ &= \lim_{\epsilon \rightarrow 0} \frac{1}{\epsilon} \left(\int_0^t h_n(s) d(\omega(s) + \epsilon g(s)) - \int_0^t h_n(s) d(\omega(s)) \right) \\ &= \lim_{\epsilon \rightarrow 0} \left(\int_0^t h_n(s) dg(s) \right) \\ &= \int_0^t h_n(s) \dot{g}(s) ds \end{aligned}$$

and from this we conclude that

$$\tilde{D}_g(F_n)(\omega) = \langle DF_n, \dot{g} \rangle_H.$$

Can we extend this to all $F_n \in \mathcal{S}$?

2. Why does

$$\frac{1}{\epsilon} (F_n \circ \tau_{\epsilon g} - F_n) = \frac{1}{\epsilon} \int_0^\epsilon \langle DF_n \circ \tau_{\alpha g}, \dot{g} \rangle_H d\alpha$$

when $F_n \in \mathcal{S}$. Above I showed that for $F = \int_0^T h(s) dB_s$ that we have

$$\frac{1}{\epsilon} (F_n \circ \tau_{\epsilon g} - F_n) = \langle h, \dot{g} \rangle_H$$

Also since

$$DF \circ \tau_{\epsilon g} = h(t) \circ \tau_{\epsilon g} = h(t)$$

because in this case DF is constant in ω then we see that

$$\langle h, \dot{g} \rangle_H = \frac{1}{\epsilon} \int_0^\epsilon \langle DF_n \circ \tau_{\alpha g}, \dot{g} \rangle_H d\alpha$$

so the statement is true for the most simple $F \in \mathcal{S}$. However I do not know how to show this for arbitrary $F \in \mathcal{S}$.

3. I don't see how Girsanov's theorem is used to conclude the convergence on the top of page 71? Also how is Girsanov's theorem used in steps 3 and 4 on page 72?

solution to 1:

The question is whether or not $F \in \mathcal{S}$ implies that the directional derivative exists and whether or not

$$\tilde{D}_g(F) = \langle DF, \dot{g} \rangle_H, \quad \text{where } g = \int \dot{g}$$

I showed above in the gray text that this is true when F takes on the most simply form of

$$F = \int_0^T h(s) dB_s$$

where $h \in H$ and B_s is the canonical brownian motion on the Weiner space. However we want this to be true for any general $F \in \mathcal{S}$ or in other words $F = f(B(h_1), \dots, B(h_n))$ where $f \in C_p^\infty(\mathbb{R}^n)$. However, such f can be uniformly approximated by functions of the form x^n so I believe that if we can show that for smooth random variables

$$F = \left(\int_0^T h(s) dB_s \right)^n$$

that

$$\tilde{D}_g(F) = \langle DF, \dot{g} \rangle$$

then we are done. However this is easily shown as follows

$$\begin{aligned} \frac{F \circ \tau_{\epsilon g} - F}{\epsilon} &= \frac{\left(\int_0^T h(s) dB_s(\omega + \epsilon g) \right)^n - \left(\int_0^T h(s) dB_s(\omega) \right)^n}{\epsilon} \\ &= \frac{\left(\int_0^T h(s) d(\omega(s)) + \epsilon \int_0^T h(s) d(g(s)) \right)^n - \left(\int_0^T h(s) d(\omega(s)) \right)^n}{\epsilon} \\ &= \frac{\sum_{i=1}^n \binom{n}{i} \epsilon^i \left(\int_0^T h(s) d(\omega(s)) \right)^{n-i} \left(\int_0^T h(s) \dot{g}(s) ds \right)^i}{\epsilon} \\ &= n \left(\int_0^T h(s) d(\omega(s)) \right)^{n-1} \langle h, \dot{g} \rangle_H + \sum_{i=2}^n \binom{n}{i} \epsilon^{i-1} \left(\int_0^T h(s) d(\omega(s)) \right)^{n-i} \left(\int_0^T h(s) \dot{g}(s) ds \right)^i \\ &\rightarrow n \left(\int_0^T h(s) d(\omega(s)) \right)^{n-1} \langle h, \dot{g} \rangle_H \quad \text{as } \epsilon \rightarrow 0 \\ &= \langle DF, \dot{g} \rangle_H \end{aligned}$$

which is what we wanted to show.

Partial solution for 2: Suppose that $F = \left(\int_0^\infty h(t) dB_t \right)^2$. Then we can calculate the following

$$F \circ \tau_{\epsilon g} - F = \left(\int_0^\infty h(t) dB_t(\omega + \epsilon g) \right)^2 - \left(\int_0^\infty h(t) dB_t(\omega) \right)^2$$

$$\begin{aligned}
&= 2\epsilon \int_0^\infty h(t)dB_t(\omega)\langle h, \dot{g} \rangle + \epsilon^2 \langle h, \dot{g} \rangle^2 \\
&= \epsilon \langle DF, \dot{g} \rangle + \epsilon^2 \langle h, \dot{g} \rangle^2
\end{aligned} \tag{4.5} \quad \boxed{\text{lhs}}$$

In addition, we can also calculate

$$\begin{aligned}
\int_0^\epsilon \langle DF \circ \tau_{\epsilon g}, \dot{g} \rangle d\alpha &= \int_0^\epsilon \int_0^\infty D_s F \circ \tau_{\epsilon g} \dot{g}(s) ds d\alpha \\
&= \int_0^\epsilon \int_0^\infty \left(2h(s) \int_0^\infty h(t) dB_t(\cdot) \right) \circ \tau_{\epsilon g} \dot{g}(s) ds d\alpha \\
&= \int_0^\epsilon \int_0^\infty \left(2 \int_0^\infty h(t) dB_t(\omega + \epsilon g) \right) h(s) \dot{g}(s) ds d\alpha \\
&= \int_0^\epsilon \int_0^\infty \left(2 \int_0^\infty h(t) dB_t(\omega) + 2\epsilon \int_0^\infty h(t) \dot{g}(t) dt \right) h(s) \dot{g}(s) ds d\alpha \\
&= \int_0^\epsilon \int_0^\infty 2 \int_0^\infty h(t) dB_t(\omega) h(s) \dot{g}(s) ds d\alpha + \int_0^\epsilon \int_0^\infty 2\epsilon \langle h, \dot{g} \rangle h(s) \dot{g}(s) ds d\alpha \\
&= \epsilon \langle DF, \dot{g} \rangle + \epsilon^2 \langle h, \dot{g} \rangle^2
\end{aligned} \tag{4.6} \quad \boxed{\text{rhs}}$$

thus we see that (4.6) and (4.5) imply that

$$\frac{1}{\epsilon} \int_0^\epsilon \langle DF \circ \tau_{\epsilon g}, \dot{g} \rangle d\alpha = \frac{1}{\epsilon} (F \circ \tau_{\epsilon g} - F)$$

and so the claim is true for $F = \left(\int_0^\infty h(t) dB_t \right)^2$ which leads us to believe that it is true for $F = \left(\int_0^\infty h(t) dB_t \right)^n$ which again leads us to believe that it is true for all $F \in \mathcal{S}$.

Informal explanation for some of Step 4: The only line that I can't quite convince myself is the first centered equation with says

$$\frac{1}{\alpha} (E((F \circ \tau_{\alpha g})G) - E(FG)) = E \left(F \frac{1}{\alpha} \left((G \circ \tau_{-\alpha g}) \exp(\alpha B(h) - \frac{1}{2} \alpha^2 \|h\|_H^2) - G \right) \right)$$

I don't understand why we pass the $\tau_{\alpha g}$ from F to G and it then becomes a $\tau_{-\alpha g}$. Also is not formally stated but the first expectation is under Q and the second is under P with $Q(A) = E(1_A L_t)$ and here

$$L_t = \exp \left(\int_0^t \dot{g}_s dB_s - \frac{1}{2} \int_0^t \dot{g}_s^2 ds \right)$$

where $\dot{g} = h$ which is stated on the previous page.

The next three centered equations are

$$\begin{aligned}
&\lim_{\alpha \rightarrow 0} \frac{1}{\alpha} \left((G \circ \tau_{-\alpha g}) \exp(\alpha B(h) - \frac{1}{2} \alpha^2 \|h\|_H^2) - G \right) \\
&= \frac{d}{d\alpha} \left((G \circ \tau_{-\alpha g}) \exp(\alpha B(h) - \frac{1}{2} \alpha^2 \|h\|_H^2) \right) \Big|_{\alpha=0}
\end{aligned} \tag{4.7}$$

$$= B(h)G - \langle DG, h \rangle_H \tag{4.8}$$

$$= \delta(Gh) \tag{4.9}$$

The first equality follows from the limit definition of the derivative. The second equality follows from products rule. We have

$$\begin{aligned}
&\frac{d}{d\alpha} \left((G \circ \tau_{-\alpha g}) \exp(\alpha B(h) - \frac{1}{2} \alpha^2 \|h\|_H^2) - G \right) \Big|_{\alpha=0} \\
&= \left(\frac{d}{d\alpha} (G \circ \tau_{-\alpha g}) \right) \exp(\alpha B(h) - \frac{1}{2} \alpha^2 \|h\|_H^2) \Big|_{\alpha=0} + (G \circ \tau_{-\alpha g}) \left(\frac{d}{d\alpha} \exp(\alpha B(h) - \frac{1}{2} \alpha^2 \|h\|_H^2) \right) \Big|_{\alpha=0}
\end{aligned}$$

Note that since $G \in \mathcal{S}$ then we have that $G = g(B(h_1), \dots, B(h_n))$ and therefore

$$G \circ \tau_{-\alpha g} = g(B(h_1), \dots, B(h_n))(\omega - \alpha g) = g(B(h_1) - \alpha \langle h_1, \dot{g} \rangle, \dots, B(h_n) - \alpha \langle h_n, \dot{g} \rangle)$$

and from this we can calculate $d/d\alpha$

$$\begin{aligned} \left. \frac{d}{d\alpha} (G \circ \tau_{-\alpha g}) \right|_{\alpha=0} &= \lim_{k \rightarrow 0} \frac{1}{k} (g(B(h_1) - (\alpha + k) \langle h_1, \dot{g} \rangle, \dots, B(h_n) - (\alpha + k) \langle h_n, \dot{g} \rangle) - G) \Big|_{\alpha=0} \\ &= -\langle DG, \dot{g} \rangle \end{aligned}$$

where the last inequality follows from the work shown in (4.3) which I believe can be applied because again $G \in \mathcal{S}$. Lastly, we simply take the derivative to see

$$\left(\frac{d}{d\alpha} \exp(\alpha B(h) - \frac{1}{2} \alpha^2 \|h\|_H^2) \right) \Big|_{\alpha=0} = B(h)$$

putting this together give us

$$\begin{aligned} \left. \frac{d}{d\alpha} \left((G \circ \tau_{-\alpha g}) \exp(\alpha B(h) - \frac{1}{2} \alpha^2 \|h\|_H^2) - G \right) \right|_{\alpha=0} &= B(h)G - \langle DF, \dot{g} \rangle_H \\ &= B(h)G - \langle DF, h \rangle_H \end{aligned}$$

which completes the verification for step 4.

Nick's Question 4.5 The formula

$$I_n(f)I_m(g) = \sum_{r=0}^{n \wedge m} r! \binom{n}{r} \binom{m}{r} I_{n+m-2r}(f \otimes_r g)$$

on page 64:

This is not a question, just wanted to test this formula out for the most basic example, for $f, g \in L^2(\mathbb{R}_+)$ and $I_1(f)I_1(g)$ so I could have an example saved.

Using the formula we would need to get

$$I_1(f)I_1(g) = I_2(f \otimes_0 g) + I_0(f \otimes_1 g) \tag{4.10}$$

$$= 2 \int_0^\infty \int_0^{t_2} f(t_1)g(t_2)dB_{t_1}dB_{t_2} + \int_0^\infty f(x)g(x)dx \tag{4.11}$$

where $(f \otimes_r g)$ is defined on page 64.

But suppose that we do the following:

$$\begin{aligned} X_t &= \int_0^t f(s)dB_s \\ Y_t &= \int_0^t g(s)dB_s \end{aligned}$$

then by applying Ito's formula we see that

$$\begin{aligned} X_t Y_t &= \int_0^t X_s dY_s + \int_0^t Y_s dX_s + \int_0^t dX_s dY_s \\ &= \int_0^t \left(\int_0^s f(\xi)dB_\xi \right) g(s)dB_s + \int_0^t \left(\int_0^s g(\xi)dB_\xi \right) f(s)dB_s + \int_0^t f(s)g(s)ds \\ &= \int_0^t \left(\int_0^s f(\xi)g(s)dB_\xi \right) dB_s + \int_0^t \left(\int_0^s g(\xi)f(s)dB_\xi \right) dB_s + \int_0^t f(s)g(s)ds \end{aligned}$$

eqn

$$\begin{aligned}
&= \int_0^t \int_0^s f(\xi)g(s) + g(\xi)f(s)dB_\xi dB_s + \int_0^t f(s)g(s)ds \\
&= \int_0^t \int_0^s 2(f\tilde{\otimes}_0 g)(\xi, s)dB_\xi dB_s + \int_0^t f(s)g(s)ds
\end{aligned}$$

where above $(f\tilde{\otimes}_0 g)$ is the symmetrization of $(f \otimes_0 g)$. Thus by letting $t \rightarrow \infty$ we see that

$$\begin{aligned}
\lim_{t \rightarrow \infty} X_t Y_t &= \int_0^\infty f(s)dB_s \int_0^\infty g(s)dB_s = I_1(f)I_1(g) \\
&= \int_0^\infty \int_0^s 2(f\tilde{\otimes}_0 g)(\xi, s)dB_\xi dB_s + \int_0^\infty f(s)g(s)ds \\
&= 2I_1(f\tilde{\otimes}_0 g) + I_0(f \otimes_1) \\
&= 2I_1(f \otimes g) + I_0(f \otimes_1)
\end{aligned} \tag{4.12}$$

cont

where the last line follows that $I_n(f) = I_n(\tilde{f})$ for any $f \in L^2(\mathbb{R}_+^n)$.

Exercises:

Exercise 2: We want to prove the following for $f, g \in L_s^2(\mathbb{R}_+^2)$

$$I_2(f)I_2(g) = I_4(f \otimes_0 g) + 4I_2(f \otimes_1 g) + 2I_0(f \otimes_2 g).$$

The hint tells us to use Ito's formula. Consider the following,

$$\begin{aligned}
X_t &= \int_0^t \int_0^\xi f(z, \xi)dB_z dB_\xi = \int_0^t F(\xi)dB_\xi \\
Y_t &= \int_0^t \int_0^\xi g(z, \xi)dB_z dB_\xi = \int_0^t G(\xi)dB_\xi
\end{aligned}$$

where we let

$$F(\xi) = \int_0^\xi f(z, \xi)dB_z \quad G(\xi) = \int_0^\xi g(z, \xi)dB_z$$

and so

$$\lim_{t \rightarrow \infty} X_t Y_t = I_2(f)I_2(g).$$

If we apply Ito's formula, we see that

$$X_t Y_t = \int_0^t X_s dY_s + \int_0^t Y_s dX_s + \int_0^t dX_s dY_s := L_1 + L_2 + L_3 \tag{4.13}$$

ito1

If this is correct, then what does dX_s and dY_s equal? For, example, is $dX_s = F(s)dB_s = \left(\int_0^s f(z, s)dB_z\right)dB_s$? If this is right, then we can continue with (4.13).

We will just look at L_1 right now.

$$\begin{aligned}
L_1 &= \int_0^t X_s dY_s \\
&= \int_0^t \underbrace{\left(\int_0^s F(\xi)dB_\xi\right)}_{X_s} G(s)dB_s
\end{aligned}$$

notice the integrand is a product of two processes. The only way I think I can continue is to apply Ito's formula again. But the problem gets messy here.

$$X_s G_s = \int_0^s X_k dG_k + \int_0^s G_k dX_k + \int_0^s dX_k dG_k$$

To continue, we need to calculate dG_k which can be done with the following Stochastic Leibniz Rule Theorem which is found [here](#). However I believe we need to assume that g is differentiable in addition to being square integrable. According to Leibniz rule we have that

$$dG_k = \left(\int_0^k \frac{\partial g}{\partial k}(z, k) dB_z \right) dk + g(k, k) dB_k$$

I dont think continuing with this strategy gets us to the desired answer though.

5 Chapter 5:

Nick's Question 5.1 Proposition 5.1.2

1. What does $F(e^{-t}B + \sqrt{1 - e^{-2t}}B')$ mean? What are B and B' ? Why is he dropping the subscripts B_t and B'_t ? I thought this was only notation but it seems that something else is happening as well? It seems like F is being evaluated at $e^{-t}B + \sqrt{1 - e^{-2t}}B'$ but this can't be possible since $e^{-t}B + \sqrt{1 - e^{-2t}}B' \in \mathbb{R}$ and $F \in L^2(\Omega)$. David Naulart [has this pdf](#). Page 5 and page 20 have some info.

2. How does

$$E'(|F(e^{-t}B + \sqrt{1 - e^{-2t}}B')|^p) = E|F|^p?$$

3. Where does

$$E' \left(\exp \left(e^{-t} \lambda B(h) + \sqrt{1 - e^{-2t}} \lambda B'(h) - \frac{1}{2} \lambda^2 \right) \right)$$

come from?

Nick's Question 5.2

Why does Mehler's formula imply that the operator T_t is nonnegative?

Nick's Question 5.3 Theorem 5.1.3

There is a [reference here](#) however it is in French.

1. Why is $|T_t F| \leq |F|$? All I think that we can say is that

$$|T_t F| \leq \sum_n e^{-nt} |I_n(f_n)| \leq \sum_n |I_n(f_n)|$$

and

$$|F| \leq \sum_n |I_n(f_n)|$$

but this does not mean that

$$|T_t(F)| \leq |F|.$$

2. Why does $|T_t F| \leq |F|$ imply that we may assume that F and G are nonnegative?
3. I think there is some typo in the claim that

$$E(\tilde{E}(F(\hat{B}))G) = E(F(\hat{B})G)$$

This is because $E(\tilde{E}(F(\hat{B}))G) \in \mathbb{R}$ while $E(F(\hat{B})G) \in \mathbb{R}^\Omega$. I am not sure what the correction should be.

Nick's Question 5.4 Lemma 5.14

1. The first line uses Stroock's formula however this can only be used if $F \in \mathbb{D}^{\infty,2}$ and in this lemma we only assume that $F \in \mathbb{D}^{k,2}$.

Nick's Question 5.5 Theorem 5.3.1

1. On the top of page 81 how do we get

$$E(\langle u, DG, \rangle_H) = - \int_0^\infty E(\langle LT_t u, DG \rangle_H) dt?$$

2. How can we show this

$$\tilde{E} \left((D^2 G)(e^{-t}B + \sqrt{1 - e^{-2t}}\tilde{B}) \right) = \frac{1}{1 - e^{-2t}} \tilde{E} \left(\tilde{D}^2 (G(e^{-t}B + \sqrt{1 - e^{-2t}}\tilde{B})) \right)$$

This is how I am interpreting this. I am guessing that $G(e^{-t}B + \sqrt{1 - e^{-2t}}\tilde{B})$ has the following wiener chaos expansion

$$G = \sum_{n=0}^{\infty} I_n(f_n)$$

where we assume that $\hat{B} = e^{-t}B + \sqrt{1 - e^{-2t}}\tilde{B}$ which means

$$\begin{aligned} I_n(f_n) &= \int_0^\infty \cdots \int_0^{z_2} f(z_1, \dots, z_n) d\hat{B}_{z_1} \dots d\hat{B}_{z_n} \\ &= \int_0^\infty \cdots \int_0^{z_2} f(z_1, \dots, z_n) d(e^{-t}B_{z_1} + \sqrt{1 - e^{-2t}}\tilde{B}_{z_1}) \dots d(e^{-t}B_{z_n} + \sqrt{1 - e^{-2t}}\tilde{B}_{z_n}) \end{aligned}$$

but something does not seem right about this.

Solution for 1 So we have that $u \in \mathbb{D}^{1,p}(H)$ and so $u = u_s(\omega)$. I believe we can say that u has Weiner Expansion

$$u_s = \sum_{n=0}^{\infty} I_n(f_{n,s})$$

with $f_{n,s} \in L_s^2(\mathbb{R}^n)$. Therefore we have

$$T_t(u_s) = \sum_{n=0}^{\infty} e^{-nt} I_n(f_{n,s})$$

and in return we then have using the formula for L on page 78

$$\begin{aligned} LT_t(u_s) &= \sum_{n=1}^{\infty} -ne^{-nt} I_n(f_{n,s}) \\ &= \sum_{n=1}^{\infty} \frac{d}{dt} e^{-nt} I_n(f_{n,s}) \end{aligned}$$

Now we integrate to see

$$\int_0^\infty LT_t(u_s) dt = \int_0^\infty \sum_{n=1}^{\infty} \frac{d}{dt} e^{-nt} I_n(f_{n,s}) dt$$

$$\begin{aligned}
&= \sum_{n=1}^{\infty} I_n(f_{n,s})[e^{-nt}]_0^{\infty} \\
&= - \sum_{n=1}^{\infty} I_n(f_{n,s}) = -u_s
\end{aligned}$$

Nick's Question 5.6 Theorem 5.5.1 Why are we able to assume that

$$\varphi(0) := \int_0^{\infty} xp(x)dx = \frac{1}{2}E|F|?$$

This assumes that ρ is a symmetric across the y-axis but this was never assumed to be the case?

6 Chapter 6

Nick's Question 6.1 page 88 How can we show

$$E \left(1_{\{\sup_{t \leq s \leq 1} B_s - B_t \geq \sup_{0 \leq s \leq t} B_s - B_t\}} | \mathcal{F}_t \right) = 2 - 2\Phi \left(\frac{\sup_{0 \leq s \leq t} (B_s - B_t)}{\sqrt{1-t}} \right)$$

The reflection principle states that for a Brownian motion, B_t , and if we define $M_t = \sup_{0 \leq s \leq t} B_s$ then

$$P(M_t \geq a) = 2P(B_t \geq a) = 2 - 2\Phi(a/\sqrt{t})$$

thus above it is like we are letting

$$a := \sup_{0 \leq s \leq t} B_s - B_t$$

and

$$t := \sqrt{1-t}$$

but this seems to me that Naulart is applying a generalization of the reflection principle that is not stated in the text and I could not find a reference for.

Nick's Question 6.2 Example 6.1.4

1. Why does Tanaka's formula imply that $\sup_{x \in \mathbb{R}} \|L_t^x\|_p < \infty$?
2. Why does

$$E(p'_\epsilon(B_s - B_r + B_r - x | \mathcal{F}_r)) = p'_\epsilon(B_r - x)$$

Nick's Question 6.3 page 91 How can we show

$$\alpha_t = \int_{\mathbb{R}} \left(\int_0^t \delta_x(B_s) ds \right)^2 dx = 2 \int_0^t \int_0^v \delta_0(B_v - B_u) du dv$$

Solution: I believe we have the following

$$\int_{\mathbb{R}} (L_t^x)^2 dx = \int_{\mathbb{R}} L_t^x L_t^x dx$$

$$\begin{aligned}
&= \int_0^t L_t^{B_s} ds \\
&= \int_0^t \int_0^t \delta_0(B_v - B_s) dv ds \\
&= 2 \int_0^t \int_0^v \delta_0(B_v - B_s) dv ds
\end{aligned}$$

where I think the last equality is by some sort of symmetry argument and the occupation formula, equation (2.17) in the book, is used in second equality.

Nick's Question 6.4 Use of backward integral on page 94 I believe that there are a series of typos that luckily do not affect the proof. First, $\{B_r - B_s : 0 \leq s \leq r\}$ is not a Brownian motion and I believe he wants to say that $\{\hat{B}_s := B_r - B_{r-s} : 0 \leq s \leq r\}$ which is a brownian motion.

Next, Naulart mentions that he is calculating $L_r^{B_r-y}$ and $L_r^{B_r+y}$ for $\hat{B}$ the reason he can do this is because

$$\begin{aligned}
L_r^{B_r-y} &= \int_0^r \delta_0(B_s - B_r + y) ds \\
&= \int_0^r \delta_0(B_r - B_s - y) ds \\
&= \int_0^r \delta_0(B_r - B_{r-s} - y) ds \\
&= \int_0^r \delta_0(\hat{B}_s - y) ds \\
&= L_r^y(\hat{B})
\end{aligned}$$

where $L_r^y(\hat{B})$ is the local time with respect to $\hat{B}$ which we will denote as L_r^y . Similarly we can show that $L_r^{B_r-y} = L_r^{-y}(\hat{B})$ which we will again denote as L_r^{-y} .

Now we apply Tanaka's formula for $\hat{B}$ to get the following

$$\begin{aligned}
L_t^y &= 2(\hat{B}_r - y)_- - 2(-y)_- + 2 \int_0^r 1_{\{\hat{B}_s < y\}} d\hat{B}_s \\
&= 2(B_r - y)_- - 2y + 2 \int_0^r 1_{\{\hat{B}_s < y\}} d\hat{B}_s
\end{aligned}$$

where we use the fact that $y > 0$ and $\hat{B}_r = B_r$. Now we apply Tanaka's formula again to see

$$\begin{aligned}
L_t^{-y} &= 2(\hat{B}_r + y)_+ - (y)_+ - 2 \int_0^r 1_{\{\hat{B}_s > -y\}} d\hat{B}_s \\
&= 2(B_r + y)_+ - 2y - 2 \int_0^r 1_{\{\hat{B}_s > -y\}} d\hat{B}_s
\end{aligned}$$

Therefore

$$\begin{aligned}
L_r^{B_r-y} - L_r^{B_r+y} &= L_r^y - L_r^{-y} \\
&= 2(B_r - y)_- - 2y + 2 \int_0^r 1_{\{\hat{B}_s < y\}} d\hat{B}_s - \left(2(B_r + y)_+ - 2y - 2 \int_0^r 1_{\{\hat{B}_s > -y\}} d\hat{B}_s \right) \\
&= 2(B_r - y)_- - 2(B_r + y)_+ + 2 \int_0^r 1_{\{|\hat{B}_s| < y\}} d\hat{B}_s
\end{aligned}$$

$$\begin{aligned}
&= 2(B_r - y)_- - 2(B_r + y)_+ + 2 \int_0^r 1_{\{|\hat{B}_{r-s}| < y\}} \hat{d}B_s \\
&= 2(B_r - y)_- - 2(B_r + y)_+ + 2 \int_0^r 1_{\{|B_r - B_s| < y\}} \hat{d}B_s
\end{aligned}$$

where we use the property of the backward integral that

$$\int_0^T u_s \hat{d}B_s = \int_0^T u_{T-s} dB_s$$

and also that $\hat{B}_{r-s} = B_r - B_s$.

Therefore we have that

$$\begin{aligned}
\Psi_h(r) &= \int_0^h L_r^{B_r-y} - L_r^{B_r+y} dy \\
&= \int_0^h L_r^y - L_r^{-y} dy \\
&= \int_0^h (2(B_r - y)_- - 2(B_r + y)_+) dy + 2 \int_0^r (h - |B_r - B_s|)_+ \hat{d}B_s
\end{aligned}$$

This is different than on page 94 but the $\int dy$ integral wont matter because $h \rightarrow 0$ and the $\int \hat{d}B_s$ integral is the same.

Nick's Question 6.5 Not sure how the conclusion of the proof of (6.8) proves what we want to show.

Nick's Question 6.6 pg 100

Why does

$$E(\delta_0(B_s^H - B_t^H)) = (2\pi)^{-d/2} |t - s|^{-Hd}$$

Solution for the case when $H = \frac{1}{2}$ and $d = 1$ (Brownian Motion)

$$\begin{aligned}
E\delta_0(B_s - B_t) &= \lim_{\epsilon \rightarrow 0} E \left[\frac{1}{2\pi\epsilon} \exp \left(-\frac{(B_s - B_t)^2}{2\pi\epsilon} \right) \right] \\
&= \lim_{\epsilon \rightarrow 0} \frac{1}{\sqrt{2\pi(\epsilon + |s - t|)}} \\
&= \frac{|s - t|^{-1/2}}{\sqrt{2\pi}}
\end{aligned}$$

where above we just calculate the expectation since $B_s - B_t \sim N(0, s - t)$. I am assuming there is some type of uniform convergence so we can exchange the limit and expected value?

7 Chapter 7

Nick's Question 7.1 Typos in Prop 7.1.2 I am pretty sure there are several typos in the proof of Prop 7.1.2. We are considering all the measurable $g : (-1, 1) \rightarrow [0, 1]$ such that $\int_{-1}^1 g(y) dy = 0$ but this means that $g = 0$ almost every where on $(-1, 1)$ which does

not seem like that is what we want. Also we then consider

$$\psi(x) = \int_{-\infty}^x g_n(y) dy$$

which does not exist and I think needs to be

$$\psi(x) = \int_{-1}^x g_n(y) dy$$

however then the claim is made that

$$\psi(F) = \int_{-1}^F g_n(y) dy \rightarrow 0$$

as $n \rightarrow \infty$ because $\int_{-1}^1 g(y) dy = 0$ but this can only happen when $F \equiv 1$ which isn't always the case.

Nick's Question 7.2 pg 107 Why can we replace $1_{F>x}$ with $1_{F<x}$ in (7.1) on page 106? It says its because the divergence has 0 expectation but I can't see why that implies the above

We have the following

$$\begin{aligned} 0 &= E \left(\delta \left(\frac{DF}{\|DF\|_H^2} \right) \right) \\ &= E \left((1_{\{F>x\}} + 1_{\{F\leq x\}}) \delta \left(\frac{DF}{\|DF\|_H^2} \right) \right) \end{aligned}$$

and therefore we must have

$$E \left(1_{\{F>x\}} \delta \left(\frac{DF}{\|DF\|_H^2} \right) \right) = -E \left(1_{\{F\leq x\}} \delta \left(\frac{DF}{\|DF\|_H^2} \right) \right)$$

which means we cant just replace $1_{\{F>x\}}$ with $1_{\{F<x\}}$

Nick's Question 7.3

1. Throughout all the existence of density results, it is claimed that the densities are bounded. Isn't this always the case? A density needs to satisfy

$$\int p(x) dx = 1$$

and so p must be bounded almost everywhere and if p is continuous then p is bounded everywhere.

2. Also why do we have that the density in Prop 7.2.5 or Theorem 7.3.1 is continuous?

solution to 2: For the case of prop 7.2.5 I believe that

$$p(x) = E \left(1_{[F>x]} H_\alpha(F, 1) \right)$$

is continuous because $H_\alpha(F, 1) \in \mathbb{D}_\infty$ and F has components that are all integrable and so I believe that it would be a easy result to show that

$$x \mapsto \int_{\Omega} 1_{[F > x]}(\omega) H_\alpha(F, 1)(\omega) dP(\omega)$$

is a continuous function. This is sort of the same thing as the standard result in real analysis that when we have a function $f \in L^1(\mathbb{R})$ then

$$x \mapsto \int_{\mathbb{R}} 1_{(-\infty, x]}(t) f(t) dt$$

is continuous.

Nick's Question 7.4 page 111 What is the definition of

$$\nabla \varphi * \nabla Q_m(x)$$

since both $\nabla \varphi$ and ∇Q_m are vectors I do not know that it means to take the convolution between them.

Also is there a subscript missing on the $\partial \varphi$ in

$$\sum_{i=1}^m \int_{\mathbb{R}^m} \partial \varphi(x - y) \partial_i Q_m(y) dy$$

and if so what should it be? On the next page it says

$$\sum_{i=1}^m \int_{\mathbb{R}^m} \partial_i \varphi(x - y) \partial_i Q_m(y) dy$$

Nick's Question 7.5 There are some detail from the proof of Lemma 7.3.2 that are confusing to me.

1. First why are we assuming p is bounded? Theorem 7.3.1 already says that p is bounded and since that is the case isn't it trivial that there exists a c such that

$$\|p\|_\infty \leq c \left(\max_{1 \leq i \leq m} \|H_{(i)}(F, 1)\|_p \right)^m$$

or is this c above then dependent on F and we do not want that. But either way, what does it mean to assume p to be bounded or not to be bounded? Why would p not be bounded if theorem 7.3.1 already says it is bounded?

2. Why do we have

$$\int_{|z-x| \leq \epsilon} |z-x|^{(1-m)q} p(x) dx \leq C_{m,p} \epsilon^{(p-m)/(p-1)} M$$

since they are pulling out the bound M we need to show that

$$\int_{|z-x| \leq \epsilon} |z-x|^{(1-m)q} dx \leq C_{m,p} \epsilon^{(p-m)/(p-1)}$$

but I don't see why this is going to be true because it seems possible that $(m-1)q \geq m$ which would mean that the integral is infinity. But since it seems that we are

arbitrarily choosing p and q we can just choose $q \leq m/(m-1)$ which avoids this problem. However I still can't figure out how they get $C_{m,p}\epsilon^{(p-m)/(p-1)}$.

solution to 2 By polar coordinates and $q = p/(p-1)$ we can say that

$$\begin{aligned} \int_{|z-x| \leq \epsilon} |z-x|^{(1-m)q} dx &= \int_{\mathcal{S}_{m-1}} \int_0^\epsilon r^{\frac{p(1-m)}{p-1}} r^{m-1} dr d\sigma \\ &= \int_{\mathcal{S}_{m-1}} \int_0^\epsilon r^{\frac{1-m}{p-1}} dr d\sigma \\ &= \epsilon^{\frac{p-m}{p-1}} \int_{\mathcal{S}_{d-1}} \frac{p-1}{p-m} d\sigma \\ &= \epsilon^{\frac{p-m}{p-1}} \sigma(\mathcal{S}_{m-1}) \frac{p-1}{p-m} \\ &= \epsilon^{\frac{p-m}{p-1}} C_{m,p} \end{aligned}$$

with $C_{m,p} = \sigma(\mathcal{S}_{m-1}) \frac{p-1}{p-m}$

Nick's Question 7.6 typo on page 114 (covering vector field) There is a typo on the top of page 114. It says that the covering vector field for F is $u = \gamma_F DF$ but it needs to say $u = \gamma_F^{-1} DF$. For example, in the $d = 2$ case we have the following

- $DF = (DF^1, DF^2)$
- $D(\varphi(F)) = \partial_1 \varphi(F) DF^1 + \partial_2 \varphi(F) DF^2$
- $\gamma_F = (\langle DF^i, DF^j \rangle)_{1 \leq i, j \leq 2}$
- $\gamma_F^{-1} = \frac{1}{|\gamma_F|} \begin{pmatrix} \langle DF^2, DF^2 \rangle & -\langle DF^1, DF^2 \rangle \\ -\langle DF^2, DF^1 \rangle & \langle DF^1, DF^1 \rangle \end{pmatrix}$
- $u := \gamma_F^{-1} (DF^1, DF^2)^T$

then from here it is an easy calculation to see that

$$\langle D(\varphi(F)), u_k \rangle = \partial_k \varphi(F)$$

Nick's Question 7.7 Possible typo in the book How is itos formula being used at the bottom of page 120?

possible solution and possible typo in book: It is shown in the book that for

$$X_t = X_0 + \int_0^t \sigma(X_s) dB_s + \int_0^t b(X_s) ds$$

that we have

$$D_r X_t = \sigma(X_r) + \int_r^t \sigma'(X_s) D_r(X_s) dB_s + \int_r^t b'(X_s) D_r(X_s) ds$$

and note that $D_r(X_r) = \sigma(X_r)$ for each $r \geq 0$. So we can rewrite the above as

$$D_r X_t = D_r(X_r) + \int_r^t \sigma'(X_s) D_r(X_s) dB_s + \int_r^t b'(X_s) D_r(X_s) ds \quad (7.1) \quad \boxed{\text{deq}}$$

Now I claim that for

$$Y_{r,t} = \int_r^t \sigma'(X_s) dB_s + \int_r^t b'(X_s) - \frac{1}{2}(\sigma')^2(X_s) ds$$

that $\exp(Y_{r,t})$ solve (7.1). By Itos formula we have that

$$\begin{aligned} \exp(Y_{r,t}) &= \exp(Y_{r,r}) + \int_r^t \sigma'(X_s) Y_{r,s} dB_s + \int_0^t Y_{r,s} \left(\frac{1}{2}(\sigma')^2(X_s) + b'(X_s) - \frac{1}{2}(\sigma')^2(X_s) \right) ds \\ &= \exp(Y_{r,r}) + \int_r^t \sigma'(X_s) Y_{r,s} dB_s + \int_0^t Y_{r,s} b'(X_s) ds \end{aligned}$$

and from with we see that $Y_{r,t}$ satisfies (7.1). However in the text book they say that

$$\sigma(X_t) \left(\int_r^t \sigma'(X_s) dB_s + \int_r^t b(X_s) - \frac{1}{2}(\sigma')^2(X_s) ds \right)$$

which I believe is incorrect.